

FLAWED SCIENCE

Scientists are human, just like the rest of us, and this means they sometimes make mistakes. Before they announce a new discovery to the world, they normally present their findings to a team of fellow scientists, who closely scrutinize errors. This process is called "peer review" and helps eliminate errors. Even so, scientists often disagree with one another, and this can make it difficult for nonscientists to know exactly what to believe. For example, climate scientists agree that the average surface temperature of Earth has risen, but they have offered a variety of different explanations for the causes of this global warming. While some scientists believe that global warming is caused by human pollution, others blame the natural cycle of Earth warming up and cooling down from the Sun's activity.

NEW DISCOVERIES

Don't try to predict the future. In 1900, Lord William Kelvin, a leading British physicist, said, "There is nothing new to be discovered in physics now." Only five years later, Albert Einstein published his groundbreaking "theory of relativity," which changed physics forever. Here are just a few things that no one knew about at the beginning of the 1900s: particle physics, nuclear energy, plate tectonics, space exploration, and black holes. These discoveries have fundamentally changed our knowledge of the universe. Can you imagine life without televisions, computers, antibiotics, lasers, and the Internet? Yet none of these had been invented then. One thing is for sure—there will always be something new to discover.

UNSOLVED MYSTERIES

Ancient peoples believed that Earth was supported on the back of a giant turtle standing on four elephants. That may sound crazy to us, but we can be sure that many ideas we now accept as scientific "truth" will seem just as absurd to future generations. Researchers at the CERN Large Hadron Collider deep under the ground in Switzerland hope that their experiments will throw light on the big bang that brought the universe into being. This is only one of hundreds of mysteries that science has yet to solve.

